

BobaLink Browser Extension “ Technical Specification

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1. Executive Summary

BobaLink is a cross-browser WebExtension (Manifest V3) designed to bridge the gap between torrent discovery on the public internet and private download management infrastructure. The extension detects magnet links and .torrent file references on any web page, deduplicates them by cryptographic infohash, and transmits them to a Boba orchestration server or directly to a qBitTorrent WebUI instance for download management.

The extension addresses three core operational challenges:

- 1. **Discovery-to-Download Latency:** Users currently copy magnet links manually, switch applications, and paste them into download clients. BobaLink reduces this workflow to a single click.
- 2. **Batch Operation Efficiency:** Tab groups containing multiple torrent pages can be processed as atomic units, enabling bulk ingestion without per-tab interaction.
- 3. **Operational Resilience:** An offline-aware queue with exponential backoff ensures that transient network failures or service maintenance do not result in lost torrents.

BobaLink is built atop the WXT framework in TypeScript, targets Chrome (¥88), Firefox (¥109), Opera, and Yandex Browser, and adheres to WCAG 2.1 AA accessibility standards. All stored credentials are encrypted at rest using AES-256-GCM via the Web Crypto API.

Key Metrics at a Glance

Metric	Target
Extension Bundle Size (compressed)	350 KB
Cold-start service worker	50 ms
Magnet link detection latency	5 ms per link
API call timeout	30 s
Offline queue retention	30 days
Supported browsers	4 (Chrome, Firefox, Opera, Yandex)
Functional requirements	25
Non-functional requirements	15

2. System Overview

BobaLink operates as a thin client within the browser's extension host environment. It does not perform any BitTorrent protocol operations itself; rather, it acts as an intelligent scraper and forwarding agent between the web browsing context and downstream download services.

2.1 High-Level Architecture

flowchart LR

```
subgraph Browser["Browser Environment"]
    A[Content Script] -->|"detected torrents"| B[Service Worker]
    C[Options Page] -->|"configure"| B
    D[Popup UI] -->|"query status"| B
    E[Context Menu] -->|"send torrent"| B
end

subgraph Network["Network Layer"]
    B -->|"HTTPS / SSE"| F[Boba Server]
    B -->|"HTTPS"| G[qBitTorrent WebUI]
end

subgraph Storage["Local Storage"]
    B --> H[(Extension Storage)]
    B --> I[(Encrypted Credentials)]
    B --> J[(Offline Queue)]
end

F -->|"RPC"| G
```

2.2 Core Workflows

Workflow 1: Single Magnet Link Detection and Forward

- Content script observes the DOM mutation events on the active tab.
- A magnet URI is matched against the regex pattern `^magnet:\?xt=urn:btih:[a-fA-F0-9]{40}`.
- The link is parsed into a structured `TorrentInfo` object.
- The torrent is sent to the service worker via `chrome.runtime.sendMessage()`.
- The service worker enqueues the item or sends it immediately based on connectivity.
- The target API (Boba or qBitTorrent) receives the torrent and initiates download.

Workflow 2: Tab Group Batch Ingestion 1. User triggers “Send Tab Group to Boba” from the popup or context menu. 2. Service worker enumerates all tabs in the specified tab group. 3. Each tab receives a `SCAN_REQUEST` message; content scripts extract all torrents. 4. Results are aggregated, deduplicated by infohash, and transmitted as a batch.

Workflow 3: Auto-Discovery 1. On first install (or when configuration is empty), the extension sends mDNS/Bonjour-like probes to well-known local network addresses. 2. Responsive Boba instances reply with their HTTPS endpoint and version. 3. The user is presented with a ranked list; selection persists to encrypted storage.

3. Functional Requirements

3.1 Torrent Detection and Parsing

FR-001: Detect Magnet Links on Any Web Page

Attribute Value

Priority	P0 “Critical
Actor	Content Script
Trigger	DOM mutation (<code>MutationObserver</code>) or page load

The extension SHALL detect all magnet URI links (`href` attributes beginning with `magnet:?`) present in the DOM of any web page. Detection MUST occur dynamically as pages mutate via single-page application (SPA) navigation.

Acceptance Criteria: - Magnet links in dynamically injected content (e.g., React/Vue re-renders) are detected within 500 ms. - Detection works across all `http:` and `https:` origins. - Magnet links inside iframes are detected when `all_frames` is enabled.

FR-002: Detect .torrent File Links

Attribute Value

Priority	P0 “Critical
Actor	Content Script
Trigger	Link pattern matching or header sniffing

The extension SHALL detect hyperlinks to `.torrent` files by: - URL path suffix matching (`.torrent` case-insensitive) - `Content-Type: application/x-bittorrent` response header (when accessible via CORS or Fetch API)

FR-003: Parse Magnet URI Parameters

Attribute Value

Priority	P0 “Critical
Actor	Parser Module

The extension SHALL parse the following magnet URI parameters per BEP-0009:

Parameter Name	Description
<code>xt</code>	Exact Topic <code>urn:btih:</code> followed by 40-char hex infohash
<code>dn</code>	Display Name Human-readable torrent name
<code>tr</code>	Tracker URL Announce URL(s); may be repeated
<code>xl</code>	Exact Length Total size in bytes (optional)
<code>ws</code>	Web Seed Web seed URL(s) (optional)
<code>x.pe</code>	Peer Address Known peer <code>host:port</code> (optional)

Validation Rules: - `xt` MUST be present and contain a valid 40-character hexadecimal infohash.
- `dn` SHOULD be present; if absent, fallback to "Unknown Torrent". - `tr` entries MUST be valid HTTP(S) URLs; malformed trackers are logged and discarded.

FR-004: Download and Parse .torrent Files

Attribute Value

Priority P1 â€” High
Actor Service Worker
Trigger `.torrent` link detection or user action

The extension SHALL download `.torrent` files via `fetch()` with the following constraints:
- Maximum file size: 10 MB (configurable via `maxTorrentFileSize` option). - Timeout: 30 seconds.
- CORS policy: Falls back to proxying via Boba server when direct fetch is blocked.

The downloaded file SHALL be parsed using a Bencode decoder to extract: - `info.name` â€” torrent name - `info.piece length` â€” piece size - `info.length` (or `info.files` for multi-file) â€” total size - `announce` and `announce-list` â€” tracker URLs

FR-005: Compute Infohash from Torrent Data

Attribute Value

Priority P1 â€” High
Actor Crypto Module

For `.torrent` files, the extension SHALL compute the SHA-1 hash over the Bencode-encoded `info` dictionary to produce the 40-character hex infohash. This MUST match the `xt` parameter if both are available.

```
// Pseudocode for infohash computation
const infoBencoded = bencode.encode(torrent.info);
const infohashBuffer = await crypto.subtle.digest('SHA-1', infoBencoded);
const infohash = hexEncode(infohashBuffer); // 40-char lowercase hex
```

FR-006: Deduplicate Torrents by Infohash

Attribute Value

Priority P1 â€” High
Actor State Manager

The extension SHALL maintain an in-memory Map<string, TorrentInfo> keyed by lowercase infohash. Duplicates (same infohash, different source page) are silently merged, preserving: - The most descriptive displayName (longest string). - The union of all tracker URLs from all sources. - The earliest detectedAt timestamp.

3.2 Integration and Transmission

FR-007: Send Magnet Links to qBitTorrent via WebUI API

Attribute Value

Priority P0 â€” Critical

Actor API Client Module

The extension SHALL transmit magnet URIs to qBitTorrentâ€™s /api/v2/torrents/add endpoint with support for: - urls parameter containing one or more magnet URIs (newline-separated). - Optional category, tags, savepath, and rename parameters. - Automatic cookie-based session management.

FR-008: Upload .torrent Files to qBitTorrent

Attribute Value

Priority P1 â€” High

Actor API Client Module

The extension SHALL upload .torrent files via multipart/form-data POST to /api/v2/torrents/add using the torrents file field. File blobs MUST be preserved in memory only for the duration of the upload request.

FR-009: Support Tab Group Batch Operations

Attribute Value

Priority P1 â€” High

Actor Tab Manager Module

The extension SHALL expose a batch-send feature that: - Enumerates all tabs within a selected Chrome tab group. - Sends a SCAN_REQUEST to each tabâ€™s content script. - Aggregates SCAN_RESPONSE results. - Deduplicates across tabs. - Transmits the unified list as a batch to the configured API.

FR-010: Enumerate Tab Groups and Extract URLs

Attribute Value

Priority P1 â€” High

Actor Tab Manager Module

Using chrome.tabGroups.query() and chrome.tabs.query(), the extension SHALL: 1. List all tab groups in the current window. 2. Display group titles and colors in the popup UI. 3. Allow the user to select one or more groups for batch operations.

3.3 Service Discovery and Authentication

FR-011: Auto-Discover Boba Services on Local Network

Attribute Value

Priority P2 â€” Medium

Actor Discovery Module

The extension SHALL attempt auto-discovery when no server is configured: - Probe `https://boba.local:8443/health`, `https://boba.local:8080/health`, and `https://boba:8443/health`. - Issue `fetch()` requests with mode: `'no-cors'` and 5-second timeout. - Parse version response; validate minimum supported version ($\geq 1.0.0$). - Present discovered endpoints in a ranked list (fastest response first).

FR-012: Support Multiple Authentication Methods

Attribute Value

Priority P1 â€” High

Actor Auth Module

The extension SHALL support the following authentication flows:

Method	Description	Default
Cookie-based	Automatic SID cookie from qBitTorrent login	Default for qBT direct
API Key	X-API-Key header for Boba server	Default for Boba mode
Basic Auth	Authorization: Basic <base64>	Optional
Custom Header	User-defined header name and value	Optional

FR-013: Encrypt Stored Credentials

Attribute Value

Priority P0 â€” Critical

Actor Crypto Module

All sensitive data persisted to `chrome.storage.local` SHALL be encrypted using AES-256-GCM via the Web Crypto API: - A 256-bit master key is derived from a per-installation random salt via PBKDF2. - The encryption key is stored in `chrome.storage.session` (in-memory only, cleared on browser restart). - Each credential entry receives a unique 96-bit IV. - Authentication tag (128-bit) is prepended to ciphertext.

3.4 Offline and Resilience

FR-014: Offline Queue with Retry

Attribute Value

Priority P1 "High

Actor Queue Manager Module

The extension SHALL maintain a persistent FIFO queue for torrents that fail to send: - Maximum queue size: 1,000 items (configurable). - Retry schedule: exponential backoff starting at 5s, capped at 5 minutes. - Maximum retry attempts: 5 (configurable). - Queue items are persisted to `chrome.storage.local` and survive browser restarts. - Failed items after max retries are moved to a "Dead Letter" sub-store for manual review.

FR-015: Real-time Download Progress via Badge

Attribute Value

Priority P2 "Medium

Actor Badge Manager Module

The extension toolbar icon badge SHALL display: - A numeric count of torrents queued for sending. - Color coding: green (idle), blue (sending), orange (queued/retrying), red (error). - Updates MUST occur within 200 ms of state changes.

3.5 User Interface

FR-016: Context Menu Integration

Attribute Value

Priority P1 "High

Actor UI Controller

The extension SHALL register the following context menu items via `chrome.contextMenus`:

Context	Label	Action
Link (magnet)	"Send Magnet to Boba"	Send detected magnet
Link (.torrent)	"Download .torrent to Boba"	Download and send file
Page	"Scan Page for Torrents"	Trigger full-page scan
Tab Group	"Send Tab Group to Boba"	Batch send tab group

FR-017: Keyboard Shortcuts

Attribute Value

Priority P2 "Medium

Actor Command Handler

The extension SHALL register the following shortcuts via `chrome.commands`:

Command	Default Binding	Action
send-current-page	Ctrl+Shift+B (Cmd+Shift+B on macOS)	Scan and send all torrents on current page
open-popup	Ctrl+Shift+L (Cmd+Shift+L on macOS)	Open extension popup
send-tab-group	Ctrl+Shift+G (Cmd+Shift+G on macOS)	Send current tab's group

All shortcuts SHALL be user-customizable via `chrome://extensions/shortcuts`.

FR-018: Cross-Browser Compatibility

Attribute Value

Priority P0 "Critical

Actor Build System

The extension SHALL target the following browsers with a single codebase:

Browser	Minimum Version	Engine	Notes
Chrome	88+	Blink	Primary development target
Firefox	109+	Gecko	Polyfill for <code>chrome.*</code> namespace
Opera	74+	Blink	Chromium-compatible
Yandex Browser	21+	Blink	Chromium-compatible

Browser-specific code SHALL be isolated behind an `BrowserAdapter` abstraction layer.

FR-019: Internationalization Support

Attribute Value

Priority P2 "Medium

Actor I18n Module

The extension SHALL support the following locales via `_locales/{lang}/messages.json`:

Locale Language

en English (default)
zh_CN Simplified Chinese
zh_TW Traditional Chinese
es Spanish
fr French
de German
ja Japanese
ko Korean

All user-facing strings MUST be externalized; no hardcoded UI text.

FR-020: Accessibility Compliance (WCAG 2.1 AA)

Attribute Value

Priority P1 "High

Actor UI Layer

The extension UI SHALL meet WCAG 2.1 Level AA: - Minimum contrast ratio 4.5:1 for normal text, 3:1 for large text. - All interactive elements keyboard-navigable (Tab order, Enter/Space activation). - ARIA labels on all icon-only buttons. - Screen reader announcements for status changes (aria-live regions). - Focus indicators visible on all focusable elements.

FR-021: Dark/Light Theme

Attribute Value

Priority P2 "Medium

Actor UI Layer

The extension SHALL respect the system's preferred color scheme via `prefers-color-scheme` and provide a manual override toggle in Options. CSS custom properties SHALL define all theme-aware colors.

FR-022: Notifications for Download Events

Attribute Value

Priority P2 "Medium

Actor Notification Manager

The extension SHALL display browser notifications via `chrome.notifications` for:

Event	Title	Body
Send success	"Torrent Sent"	{name} added to download queue
Send failed	"Send Failed"	{name} "click to retry
Batch complete	"Batch Complete"	{count} torrents sent, {failed} failed
Queue retry	"Retrying Send"	Attempt {n} for {name}

Notifications SHALL be suppressible per-category in Options.

FR-023: Options Page Configuration

Attribute Value

Priority P1 "High

Actor Options Page

The Options page SHALL provide configuration for: - Server URL and connection mode (Boba proxy / qBitTorrent direct). - Authentication credentials (with masked input and test-connection button). - Default category, tags, and download path. - Notification preferences. - Queue settings (max size, retry count, backoff strategy). - Keyboard shortcut display (non-editable; links to

browser shortcuts page). - Theme selection (system / light / dark). - Reset to defaults with confirmation.

FR-024: Health Check and Connection Status

Attribute Value

Priority P1 â€” High

Actor Health Monitor

The extension SHALL display a persistent connection status indicator: - Polling interval: 30 seconds (configurable). - Endpoint: `/api/v2/app/version` for qBitTorrent; `/health` for Boba. - Status states: `connected`, `connecting`, `disconnected`, `error`. - Visual indicator in popup header; tooltip shows last successful ping time.

FR-025: Rate Limiting for API Calls

Attribute Value

Priority P1 â€” High

Actor API Client Module

The extension SHALL enforce client-side rate limiting: - Default: 10 requests per second burst, 60 requests per minute sustained. - Queue processing: maximum 1 concurrent API call with 500 ms inter-call delay. - Respects HTTP 429 Retry-After header from server. - All rate-limit state is held in-memory (non-persistent).

4. Non-Functional Requirements

4.1 Performance

ID	Requirement	Target	Measurement Method
NFR-001	Content script initialization	â‰¤ 10 ms	<code>performance.now()</code> delta
NFR-002	Magnet link detection (per link)	â‰¤ 5 ms	Benchmark on 1,000-link page
NFR-003	Popup render time	â‰¤ 100 ms	Lighthouse performance audit
NFR-004	Options page load time	â‰¤ 200 ms	Lighthouse performance audit
NFR-005	API call round-trip (local network)	â‰¤ 500 ms	95th percentile over 100 calls
NFR-006	Service worker cold start	â‰¤ 50 ms	Chrome DevTools Performance tab
NFR-007	Bundle size (compressed CRX)	â‰¤ 350 KB	<code>du -h</code> on build artifact

4.2 Security

ID	Requirement	Implementation
NFR-008	Credential encryption at rest	AES-256-GCM via Web Crypto API
NFR-009	No cleartext credential logging	ESLint rule + code review
NFR-010	HTTPS-only API communication	URL scheme validation + CSP
NFR-011	Content Security Policy	<code>script-src 'self';</code> <code>object-src 'none'</code>
NFR-012	Minimum permission model	<code>activeTab</code> + declared host permissions only

4.3 Reliability

ID	Requirement	Target
NFR-013	Crash-free session rate	≥ 99.9%
NFR-014	Offline queue durability	100% – all items survive restart
NFR-015	API compatibility coverage	qBitTorrent 4.4.x through 5.x

4.4 Scalability

- Concurrent tab group batch size: unlimited (memory-bound).
- Offline queue: 1,000 items default, 10,000 maximum.
- Detection: pages with 10,000+ links must not cause UI jank (> 16 ms frame time).

4.5 Maintainability

- Code coverage: ≥ 80% unit test coverage, ≥ 60% E2E scenario coverage.
- Documentation: all public APIs documented with JSDoc.
- TypeScript: strict mode enabled, zero any types in production code.
- Dependency count: ≤ 20 runtime dependencies.

5. System Architecture

5.1 Component Diagram

```
graph TB
    subgraph "Browser Extension Host"
        CS[Content Script]
        SW[Service Worker]
        OP[Options Page]
        PU[Popup UI]
    end

    subgraph "Core Modules (Service Worker)"
        PM[Parser Module]
        AM[Auth Module]
    end
```


5.3 Interface Definitions

Content Script ↔ Service Worker (`chrome.runtime.sendMessage`)

```
interface DetectionMessage {
  type: 'TORRENT_DETECTED';
  payload: {
    torrents: TorrentInfo[];
    pageUrl: string;
    timestamp: number;
  };
}
```

Service Worker ↔ Content Script (`chrome.tabs.sendMessage`)

```
interface ScanRequestMessage {
  type: 'SCAN_REQUEST';
  payload: {
    requestId: string;
    deepScan: boolean;
  };
}

interface ScanResponseMessage {
  type: 'SCAN_RESPONSE';
  payload: {
    requestId: string;
    torrents: TorrentInfo[];
    error?: string;
  };
}
```

6. Data Models

6.1 TorrentInfo

```
interface TorrentInfo {
  /** 40-character lowercase hex infohash */
  infohash: string;

  /** Human-readable torrent name */
  displayName: string;

  /** Magnet URI (reconstructed from parsed data) */
  magnetUri: string;

  /** Source type */
  source: 'magnet-link' | 'torrent-file';

  /** Announce URLs */
  trackers: string[];
}
```

```

/** Total size in bytes, if known */
totalSize?: number;

/** Web seed URLs */
webSeeds?: string[];

/** URL of the page where detected */
pageUrl: string;

/** Timestamp of detection (Unix ms) */
detectedAt: number;

/** Raw torrent file blob (for .torrent uploads only) */
fileBlob?: Blob;
}

```

6.2 ServerConfig

```

interface ServerConfig {
  /** Connection mode */
  mode: 'boba' | 'qbittorrent-direct';

  /** Base URL (e.g., https://boba.local:8443) */
  baseUrl: string;

  /** Authentication method */
  authMethod: 'cookie' | 'api-key' | 'basic' | 'custom-header';

  /** Credentials (encrypted at rest) */
  credentials: EncryptedCredentials;

  /** Default category for new torrents */
  defaultCategory?: string;

  /** Default tags for new torrents */
  defaultTags?: string[];

  /** Default save path */
  defaultSavePath?: string;

  /** Connection timeout in ms */
  timeout: number;

  /** Health check interval in ms */
  healthCheckInterval: number;
}

```

6.3 ExtensionConfig

```

interface ExtensionConfig {
  /** Extension version that wrote this config */
  version: string;
}

```

```

/** Server configuration */
server: ServerConfig;

/** UI preferences */
ui: {
  theme: 'system' | 'light' | 'dark';
  badgeEnabled: boolean;
  notifications: {
    sendSuccess: boolean;
    sendFailed: boolean;
    batchComplete: boolean;
    queueRetry: boolean;
  };
};

/** Queue behavior */
queue: {
  maxSize: number;
  maxRetries: number;
  retryBaseDelayMs: number;
  retryMaxDelayMs: number;
};

/** Detection preferences */
detection: {
  scanDynamically: boolean;
  maxTorrentFileSize: number;
  enableTabGroupScan: boolean;
};

/** Rate limiting */
rateLimit: {
  requestsPerSecond: number;
  requestsPerMinute: number;
  interCallDelayMs: number;
};
}

```

6.4 QueueItem

```

interface QueueItem {
  /** Unique queue entry ID */
  id: string;

  /** The torrent to send */
  torrent: TorrentInfo;

  /** Current status */
  status: 'pending' | 'retrying' | 'failed' | 'dead-letter';

  /** Number of send attempts made */
  attemptCount: number;
}

```

```

/** Next scheduled retry time (Unix ms) */
nextRetryAt: number;

/** Last error message */
lastError?: string;

/** HTTP status code of last failure, if applicable */
lastHttpStatus?: number;

/** When the item was first queued */
queuedAt: number;
}

```

7. API Specifications

7.1 Internal Message Passing Protocol

All inter-script communication uses `chrome.runtime.sendMessage()` and `chrome.tabs.sendMessage()`.

Message Type	Direction	Payload	Description
SCAN_REQUEST	SW → CS	{ requestId, deepScan }	Request full page scan
SCAN_RESPONSE	CS → SW	{ requestId, torrents, error? }	Scan results
TORRENT_DETECTED	CS → SW	{ torrents, pageUrl, timestamp }	Real-time detection event
SEND_TORRENT	PU/SW → ACM	{ torrent, priority? }	Queue or send immediately
SEND_BATCH	TM → ACM	{ torrents[] }	Batch send request
GET_STATUS	PU → SW	{ }	Request connection + queue status
STATUS_RESPONSE	SW → PU	{ connected, queueSize, pendingCount }	Status data
GET_QUEUE	PU → QM	{ }	Request full queue contents
QUEUE_RESPONSE	QM → PU	{ items: QueueItem[] }	Queue snapshot
RETRY_ITEM	PU → QM	{ itemId }	Manually retry a failed item
REMOVE_ITEM	PU → QM	{ itemId }	Remove item from queue
CLEAR_QUEUE	PU → QM	{ status?: string }	Clear all or filtered items

7.2 Boba API Integration

Base URL Configuration

`https://{host}:{port}/api/v1`

Authentication

POST /auth/token
Content-Type: application/json

```
{ "apiKey": "string" }
```

Response: { "token": "jwt-string", "expiresAt": 1700000000 }

Search API

GET /torrents/search?q={query}&category={category}&limit={limit}
Authorization: Bearer {token}

Response: { "results": TorrentInfo[], "total": number }

Download API

POST /torrents/download
Authorization: Bearer {token}
Content-Type: application/json

```
{ "infohash": "string", "category": "string?", "tags": ["string"] }
```

Response: { "jobId": "string", "status": "queued" }

SSE Streaming

GET /events/download-progress
Authorization: Bearer {token}
Accept: text/event-stream

Event format:
event: progress
data: { "infohash": "...", "progress": 0.45, "speed": 1024000, "eta": 360 }

7.3 qBitTorrent WebUI API

Detailed endpoint specifications are provided in [api-reference.md](#).

8. Security Architecture

8.1 Threat Model

Threat	Severity	Mitigation
Credential theft from disk	High	

Threat	Severity	Mitigation
Man-in-the-middle attack	High	AES-256-GCM encryption; key in session-only storage HTTPS-only; certificate pinning optional
XSS via malicious torrent site	Medium	CSP; content script isolation; no eval()
Privilege escalation	Medium	Minimal permission manifest; activeTab default
Extension fingerprinting	Low	No unique identifiers exposed to web pages

8.2 Encryption Flow

sequenceDiagram

```

    participant U as User
    participant E as Extension
    participant S as chrome.storage.session
    participant L as chrome.storage.local
  
```

```

    U->>E: Enter password/API key
    E->>E: Generate random salt (16 bytes)
    E->>E: Derive key via PBKDF2(100k iterations)
    E->>S: Store encryption key
    E->>E: Encrypt credential with AES-256-GCM
    E->>L: Store { iv + authTag + ciphertext, salt }
  
```

Note over E,L: On extension startup

```

    E->>S: Check for existing key
    alt Key exists
        S-->>E: Return key
    else Key missing
        E->>E: Prompt for password
        E->>E: Re-derive key from salt + password
        E->>S: Store key
    end
  
```

8.3 Content Security Policy

```

{
  "content_security_policy": {
    "extension_pages": "script-src 'self'; object-src 'none'; connect-src
  }
}
  
```

9. Error Handling Strategy

9.1 Error Categories

Code	Category	Retry Strategy	User Notification
E_NETWORK	Network failure (DNS, TCP)	Exponential backoff	Silent (badge only)
E_TIMEOUT	Request timeout	Immediate retry $\times 2$, then backoff	Silent
E_AUTH	Authentication failure	No retry; manual intervention required	Immediate notification
E_RATE_LIMIT	HTTP 429	Honor Retry-After, then backoff	Silent
E_VALIDATION	Invalid torrent data	No retry	Immediate notification
E_SERVER	5xx server error	Backoff with jitter	Badge + optional notification
E_STORAGE	Local storage full	No retry; alert user	Immediate notification

9.2 Error Object Schema

```
interface ExtensionError {
  code: 'E_NETWORK' | 'E_TIMEOUT' | 'E_AUTH' | 'E_RATE_LIMIT' | 'E_VALIDATION';
  message: string;
  timestamp: number;
  context?: Record<string, unknown>;
  recoverable: boolean;
}
```

10. Performance Considerations

10.1 Memory Management

- Content scripts use WeakMap for DOM element $\rightarrow$ torrent metadata mapping to avoid leaks.
- BLOB references for .torrent files are revoked immediately after upload via `URL.revokeObjectURL()`.
- Service worker uses lazy initialization for all modules; unused modules are not loaded.

10.2 DOM Observation Strategy

```
// Optimized MutationObserver
const observer = new MutationObserver((mutations) => {
  const hasLinkMutations = mutations.some(m =>
    Array.from(m.addedNodes).some(n =>
      n instanceof HTMLElement &&
      (n.tagName === 'A' || n.querySelector('a'))
    )
  );
  if (hasLinkMutations) {
```

```

    debouncedScan();
  }
});

observer.observe(document.body, { childList: true, subtree: true });

```

- Scanning is debounced at 250 ms to batch rapid DOM mutations.
- `requestIdleCallback` is used for non-urgent parsing on pages with > 500 links.

10.3 Bundle Optimization

- Tree-shaking enabled via WXT/Vite.
- Code-splitting: Options page and Popup are separate chunks.
- Shared vendor chunk for crypto and parsing utilities.
- LZ-string compression for queue persistence.

11. Technology Stack

Layer	Technology	Version	Purpose
Language	TypeScript	^5.0	Type-safe development
Framework	WXT	^0.17	Extension build tooling
Bundler	Vite	^5.0	Module bundling and HMR
Testing (Unit)	Jest	^29.0	Unit and integration tests
Testing (E2E)	Playwright	^1.40	Browser automation tests
Linting	ESLint	^8.0	Code quality
Formatting	Prettier	^3.0	Code formatting
Crypto	Web Crypto API	Native	AES-256-GCM, SHA-1, PBKDF2
Storage	Extension Storage API	MV3	<code>chrome.storage.local/session</code>
UI	Vanilla TS + CSS	Native	No framework dependency

11.1 Runtime Dependencies

```

bencode-js          ^3.0.0    Bencode encoder/decoder for .torrent files
lz-string            ^1.5.0    Compression for queue persistence
webextension-polyfill ^0.10.0    Firefox chrome.* namespace compatibility

```

11.2 Dev Dependencies

```

@types/chrome        ^0.0.260  Chrome extension API types
@types/jest           ^29.5.0   Jest type definitions
ts-jest              ^29.1.0   TypeScript Jest transformer
jest-environment-jsdom ^29.7.0   DOM environment for unit tests
@playwright/test      ^1.40.0   E2E testing framework

```

12. Deployment Architecture

12.1 Build Outputs

```
dist/
â”€â”€â”€â”€â”€ chrome/                # Chrome/Opera/Yandex build (MV3)
â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€ manifest.json
â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€ background.js
â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€ content-scripts/
â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€ popup/
â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€ options/
â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€ _locales/
â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€ firefox/        # Firefox build (MV3 with polyfills)
â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€ ...
â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€ edge/           # Edge build (identical to Chrome)
â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€ ...
```

12.2 CI/CD Pipeline

```
graph LR
  A[Developer Push] --> B[Lint & Type Check]
  B --> C[Unit Tests]
  C --> D[Build All Targets]
  D --> E[E2E Tests]
  E --> F[Code Coverage Gate]
  F --> G{Branch?}
  G -->|main| H[Deploy to Chrome Web Store]
  G -->|main| I[Deploy to AMO]
  G -->|main| J[Create GitHub Release]
```

12.3 Distribution Channels

Channel	Format	Update Mechanism
Chrome Web Store	CRX	Automatic
Firefox AMO	XPI	Automatic
Opera Addons	CRX	Automatic
GitHub Releases	ZIP	Manual

13. Assumptions and Dependencies

13.1 Assumptions

1. The target qBitTorrent instance has the WebUI enabled and accessible via HTTPS.
2. Users have administrative access to configure Boba server or qBitTorrent settings.
3. The local network allows HTTP(S) communication between the browser and Boba/qBitTorrent.
4. Users will grant the `activeTab` permission when prompted for site-specific operations.
5. The browser supports Manifest V3 (all targeted versions do).

13.2 External Dependencies

Dependency	Minimum Version	License
Node.js	18 LTS	MIT
qBitTorrent (WebUI)	4.4.0	GPL-2.0+
Boba Server	1.0.0	Proprietary
Chrome / Chromium	88	N/A
Firefox	109	MPL-2.0

14. Risks and Mitigations

Risk	Probability	Impact	Mitigation
Chrome policy changes restrict MV3 capabilities	Medium	High	Maintain MV2 fallback branch; participate in Chromium feature requests
qBitTorrent WebUI API breaking changes	Medium	Medium	Version detection; adapter pattern for API differences
CORS blocking torrent file downloads	High	Medium	Proxy downloads through Boba server; fallback notification
User forgets encryption password	Medium	Medium	Optional password hint; ability to reset and reconfigure
Extension store policy rejection	Low	High	Pre-submission review; compliant data handling practices
Third-party dependency vulnerability	Medium	High	Automated Dependabot alerts; lockfile pinning; SCA scanning
Large pages causing performance issues	Medium	Medium	<code>requestIdleCallback</code> ; pagination for large link sets

15. Appendices

Appendix A: Glossary

Term	Definition
Infohash	A 20-byte (40-character hex) SHA-1 hash uniquely identifying a torrent's metadata.
Magnet URI	A URI scheme (<code>magnet:</code>) containing metadata for peer-to-peer file sharing, per BEP-0009.
Bencode	The encoding format used by BitTorrent for <code>.torrent</code> files; supports dictionaries, lists, integers, and byte strings.
Boba	The orchestration server that proxies and manages torrent download requests.

Term	Definition
qBitTorrent	An open-source BitTorrent client with a built-in WebUI API.
Content Script	A script injected by the extension into web pages to read and modify the DOM.
Service Worker	The event-driven background script in MV3 that handles API calls, state management, and inter-script messaging.
MV3	Manifest V3, the third version of the WebExtension manifest format.
WXT	A modern build tool for browser extensions that wraps Vite.
mDNS	Multicast DNS, a protocol for name resolution on local networks without a dedicated DNS server.
CSP	Content Security Policy, a browser mechanism to control resource loading.
AES-256-GCM	Advanced Encryption Standard with 256-bit key in Galois/Counter Mode; provides authenticated encryption.
PBKDF2	Password-Based Key Derivation Function 2; used to derive cryptographic keys from passwords.
SSE	Server-Sent Events, a push technology for real-time updates over HTTP.
WCAG	Web Content Accessibility Guidelines, a standard for web accessibility.

Appendix B: References

1. [BitTorrent BEP-0003: The BitTorrent Protocol Specification](#)
2. [BitTorrent BEP-0009: Magnet Links](#)
3. [BitTorrent BEP-0012: Multitracker Metadata Extension](#)
4. [qBitTorrent WebUI API Documentation](#)
5. [Chrome Extension Manifest V3](#)
6. [Mozilla WebExtensions API](#)
7. [WXT Framework Documentation](#)
8. [WCAG 2.1 Specification](#)
9. [Web Crypto API Specification](#)
10. [Server-Sent Events \(MDN\)](#)

Appendix C: Document History

Version	Date	Author	Changes
0.1.0	2026-05-15	Platform Team	Initial draft
0.2.0	2026-05-28	Platform Team	Added security architecture, NFRs
0.3.0	2026-06-02	Architecture Board	Review feedback incorporated
1.0.0	2026-06-06	Platform Team	Final release

End of Technical Specification